

Image fusion meets deep learning: A survey and perspective

Hao Zhang^a, Han Xu^a, Xin Tian^a, Junjun Jiang^b, Jiayi Ma^{a,*}

^a*Electronic Information School, Wuhan University, Wuhan, 430072, China*

^b*School of Computer Science and Technology, Harbin Institute of Technology, Harbin, 150001, China*

Abstract

Image fusion, which refers to extracting and then combining the most meaningful information from different source images, aims to generate a single image that is more informative and beneficial for subsequent applications. The development of deep learning has promoted tremendous progress in image fusion, and the powerful feature extraction and reconstruction capabilities of neural networks make the fused results promising. Recently, several latest deep learning technologies have made image fusion explode, *e.g.*, generative adversarial networks, autoencoder, *etc.* However, a comprehensive review and analysis of latest deep-learning methods in different fusion scenarios is lacking. To this end and in this survey, we first introduce the concept of image fusion, and classify the methods from the perspectives of the deep architectures adopted and fusion scenarios. Then, we review the state-of-the-art on the use of deep learning in various types of image fusion scenarios, including the digital photography image fusion, the multi-modal image fusion and the sharpening fusion. Subsequently, the evaluation for some representative methods in specific fusion tasks are performed qualitatively and quantitatively. Moreover, we briefly introduce several typical applications of image fusion, including RGBT object tracking, medical diagnosis, photography visualization, and remote sensing monitoring. Finally, we provide the conclusion, highlight the challenges in image fusion, and look forward to potential future research directions.

Keywords: Image fusion, deep learning, digital photography, multi-modal, sharpening

*Corresponding author

Email addresses: zhpersonalbox@gmail.com (Hao Zhang), xu_han@whu.edu.cn (Han Xu), xin.tian@whu.edu.cn (Xin Tian), jiangjunjun@hit.edu.cn (Junjun Jiang), jyima2010@gmail.com (Jiayi Ma)

1. Introduction

Due to the theoretical and technical limitations of hardware devices, images captured by a single sensor or under a single shooting setting cannot effectively and comprehensively describe the imaging scene [1]. Naturally, image fusion is able to combine the meaningful information in different source images to generate a single image, which contains richer information and is more beneficial for subsequent applications [2]. Due to the excellent characteristics of fused images, the image fusion as an image enhancement method has been widely used in many fields, such as the photography visualization [3], object tracking [4], medical diagnosis [5], and remote sensing monitoring [6].

Image fusion has already received intensive research before the prevalence of deep learning. The early methods to realize image fusion adopt the related mathematical transformation to manually analyze the activity level and design fusion rules in the spatial domain or transformation domain, called traditional fusion methods [7, 8, 9, 10]. Typical traditional fusion methods include multi-scale transform-based methods [11], sparse representation-based methods [12], subspace-based methods [13], saliency-based methods [14], total variation-based methods [15], *etc.* However, the limitations of these methods have become increasingly apparent. On the one hand, to ensure the feasibility of subsequent feature fusion, the traditional methods are forced to adopt the same transformation for different source images to extract features. However, this operation does not consider the characteristic difference of source images, which may result in poor expressiveness of the extracted features. On the other hand, the traditional feature fusion strategies are so rough that the fusion performance is very limited. The motivation for introducing deep learning into image fusion is to overcome these limitations of traditional methods [1, 16, 17]. First, deep learning-based methods can use different network branches to fulfill differentiated feature extraction, so as to obtain more targeted features. Second, deep learning-based methods can learn a more reasonable feature fusion strategy under the guidance of well- designed loss functions, so as to realize adaptive feature fusion. Benefiting from these advantages, deep learning promotes tremendous progress in image fusion, obtaining performance far exceeding traditional methods.

From the perspective of fulfilling image fusion, the existing deep learning-based fusion meth-

ods are dedicated to solving some or all of three sub-problems in image fusion, *i.e.*, feature extraction, feature fusion and image reconstruction. In terms of the adopted deep architecture, these deep learning-based fusion strategies can be divided into autoencoder (AE)-based, convolutional neural network (CNN)-based, and generative adversarial network (GAN)-based methods. The AE methods usually pre-train an autoencoder. Then the trained autoencoder is used to achieve feature extraction and image reconstruction, while the intermediate feature fusion is realized according to conventional fusion rules [18], as shown in Fig. 1 (a). For example, DenseFuse [19] is one of the most famous AE-based methods, which trains the encoder and decoder on MSCOCO [20] dataset and adopts the addition and the l_1 -norm fusion strategies. As for the CNN methods, they usually introduce the convolutional neural network into image fusion in two different forms. One is to implement feature extraction, feature fusion and image reconstruction end-to-end by using well-designed loss functions and network structures [21], as shown in Fig. 1 (b). A representative work of the end-to-end CNN-based methods is PMGI [16], which proposes the proportional maintenance loss of gradient and intensity to guide the network to directly generate fused images. The other is to adopt a trained CNN to formulate fusion rules, while the feature extraction and image reconstruction are done using traditional methods [22], as shown in Fig. 1 (c). For instance, Liu *et al.* [23] adopted CNN to generate fusion weights, while image decomposition and reconstruction are implemented by Laplacian pyramids. The GAN methods rely on the adversarial game between the generator and the discriminator to estimate the probability distribution of the target, which can jointly fulfill feature extraction, feature fusion and image reconstruction in an implicit manner [24, 25], as shown in Fig. 1 (d). FusionGAN is the pioneer of GAN-based image fusion, which establishes an adversarial game between the fused image and the visible image to further enrich the texture details in the fused image. Because of the significant differences in various image fusion tasks, the implementation of these methods in diverse fusion scenarios is different. Therefore, it is appropriate to review the deep learning-based fusion methods according to the categories of image fusion tasks.

Typically, image fusion scenarios can be divided into three categories according to the fusion target and the differences in imaging of source images, as shown in Fig. 2. i) Digital photography image fusion. Due to the performance limitations of digital imaging devices, the sensor cannot

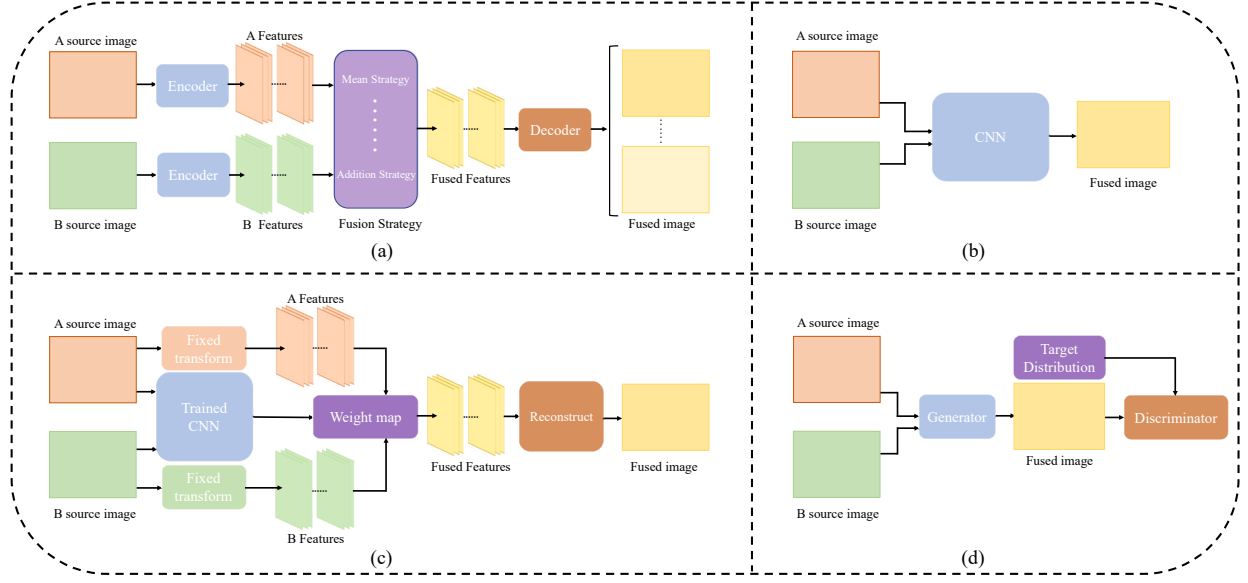


Figure 1: Different deep architectures for image fusion. (a) AE-based architecture; (b) and (c) CNN-based architectures; (d) GAN-based architecture.

fully characterize the information in imaging scenes under a single setting. For example, images produced by digital photography only withstand a limited illumination variation and have a pre-defined depth-of-field. In this context, as typical tasks of digital photography image fusion, the multi-exposure image fusion and multi-focus image fusion can merge images captured under different settings to produce results with high dynamic range and full clarity. ii) Multi-modal image fusion. Due to the limitation of imaging principles, a single sensor can only capture part of the scene information. Multi-modal image fusion combines the most significant information in the images obtained by multiple sensors to achieve an effective description of the scene. Representative multi-modal image fusion tasks include the infrared and visible image fusion and medical image fusion. iii) Sharpening fusion. Under the premise of ensuring the signal-to-noise ratio, there is a certain contradiction between spectroscopic/filter and instantaneous field of view (IFOV). In other words, no sensor can capture images with both high spatial resolution and high spectral resolution. Sharpening fusion is dedicated to fusing images of different spatial/spectral resolutions to generate desired results, which have not only high spatial resolution but also high spectral resolution. Typical sharpening fusions include multi-spectral (MS) sharpening and hyperspectral sharpening.

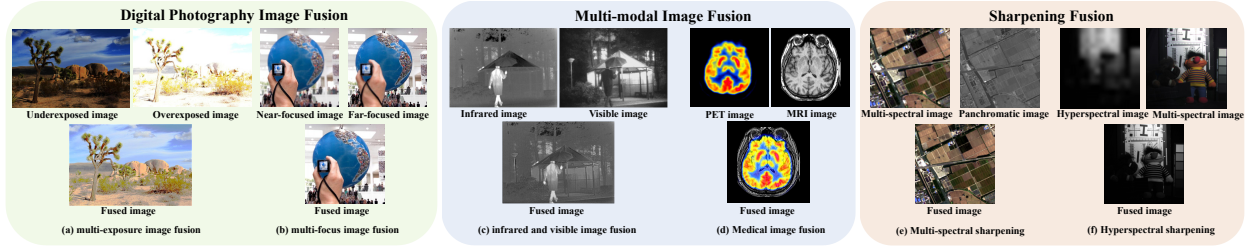


Figure 2: Schematic diagram for three categories of image fusion.

From the perspective of source image imaging, sharpening fusion also belongs to multi-modal image fusion. However, in terms of fusion targets, sharpening fusion requires higher spectral/spatial fidelity than the aforementioned multi-modal image fusion, bringing a direct resolution improvement. Therefore, the sharpening fusion is discussed as a separate category.

The motivation of this work is as follows. On the one hand, existing surveys mainly focus on a specific fusion task, and few work comprehensively reviews the latest technologies in multiple image fusion scenarios [26, 27, 28, 29, 30]. For example, [2] only investigates the development of infrared and visible image fusion methods, and [31] only provides a review of multi-focus image fusion methods. Moreover, these works focus more on traditional methods, and the survey of deep learning-based methods is not comprehensive [32, 33, 34, 35]. On the other hand, in the past few years, image fusion has developed explosively by virtue of deep learning, and the research on deep fusion methods has entered a new stage. However, the existing survey [36] has missed the latest deep fusion methods such as the GAN-based and AE-based methods. To this end, we propose a comprehensive survey for the state-of-the-art on the use of deep learning in various types of image fusion scenarios, which could help relevant researchers and technicians better understand the current development status of deep fusion methods.

The overall framework of this survey is shown in Fig. 3. In particular, the representative deep approaches are firstly discussed according to three categories of image fusion scenarios, *i.e.*, digital photography image fusion, multi-modal image fusion and sharpening fusion. Then we conduct a brief evaluation for representative deep learning-based methods in each fusion task. Subsequently, some typical applications of image fusion are introduced, *i.e.*, photography visualization, RGBT object tracking, medical diagnosis and remote sensing monitoring. Finally, we summarize this

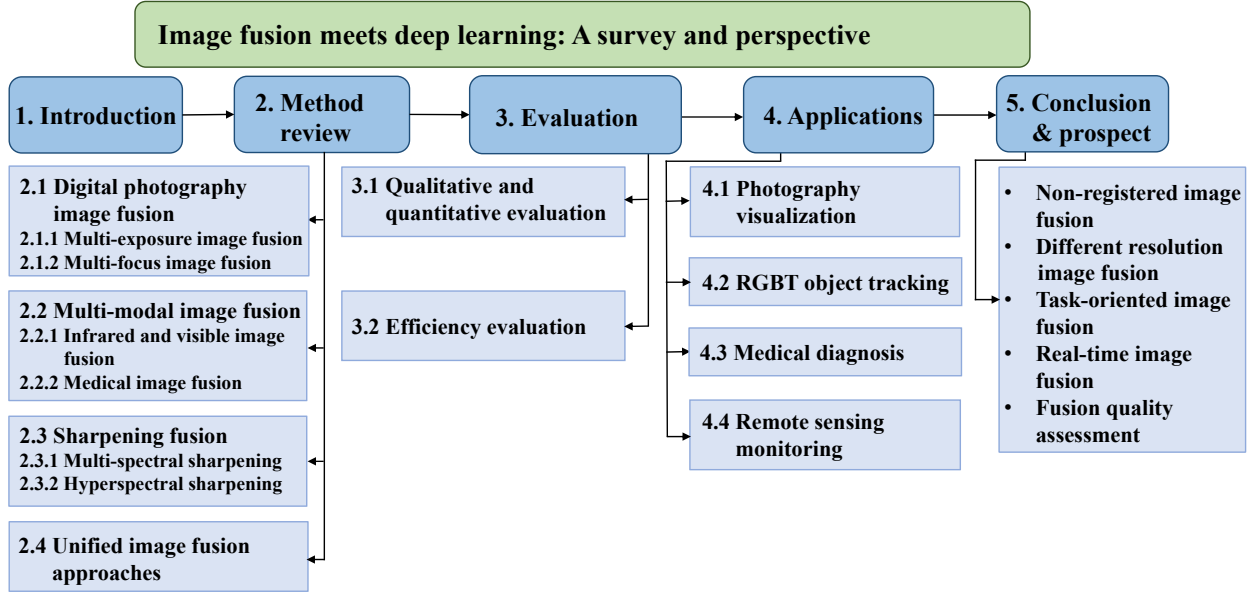


Figure 3: Overall framework of the survey.

survey, and put forward some prospects for the future study in combination with the challenges that exist in image fusion.

2. Method review

2.1. Digital photography image fusion

Digital imaging devices first use the optical lens to capture the reflected visible light, and then adopt the digital module, such as CCD and CMOS, to record the scene information. On the one hand, because the optical lens is limited by the depth-of-field, all objects usually cannot be focused at the same time. On the other hand, these digital modules cannot withstand excessive imaging exposure differences due to the limited dynamic range. Digital photography image fusion is a good choice to address these challenges, that combines multiple images captured under different shooting settings to generate full-clear images with high dynamic range. Therefore, two typical digital photography image fusion tasks are derived, saying multi-exposure image fusion and multi-focus image fusion.

2.1.1. Multi-exposure image fusion

The illumination changes between regions of the imaging scene are often large. Due to the technical limitations of imaging devices, the images captured by the sensor are usually under very limited dynamic range of exposures [37]. In other words, the images taken under a single shooting setting will lose scene information because of overexposure/underexposure. Multi-exposure image fusion is to combine the effective information of images with different exposures to produce the result with appropriate exposure and rich imaging information. The current methods to achieve multi-exposure fusion mainly include the CNN and GAN methods.

Some CNN methods directly use the trained network to extract features from source images with different exposures, and then determine the importance of the corresponding pixel position based on the salience of feature maps to generate the fusion weight map [38]. The final fused result is generated by directly weighting source images according to the obtained fusion weight map. It is worth noting that the salience of feature maps do not necessarily reflect the validity of the information in source images, which means that disappointing results may be obtained. Some other CNN methods learn directly on multi-exposure data to dynamically complete feature extraction, feature fusion and image reconstruction, which generally follow two types of technical routes. One is to select a well-exposed image as the ground truth and use it to supervise the training of the fusion network [39]. Nevertheless, this manually selected reference image is very subjective, which may set an upper limit for the learning of the fusion network. Another technical route is to use special-designed non-reference metrics, such as MEF-SSIM [40], to evaluate the quality of fused results, so as to guide the network to produce well-exposed images [21, 41, 42]. In terms of fusion performance, these methods based on no-reference metrics have produced promising fused results. Since the rationality of the non-reference metric directly determines the learning quality of networks, the key to the problem lies in the further development of high-performance non-reference metrics for evaluating the fused results.

The GAN methods take the exposure condition as a probability distribution, so that the exposure of the fused image tends to be good through adversarial learning. Then it is important to construct the ideal exposure distribution as the target of approximation. MEF-GAN [43] uses the label image in the MEF dataset as the exposure reference, which is selected from the results

of multiple image fusion algorithms. As mentioned above, this artificially selected ground truth is not the best choice. GANFuse [44] constructs the adversarial model from the perspective of information combination. Specifically, GANFuse believes that the difference between the fused image and a source image should have a similar probability distribution as another source image. Although such an adversarial model can make the fused image contain as much information as possible from the source image, it may be inaccurate to assume that the information in the fused image is the sum of that in source images. Nevertheless, the GAN methods still produce promising fused results. From this point of view, making full use of the information (*e.g.*, exposure condition, scene structure) in source images to establish the unsupervised adversarial model may be a good choice to achieve high-quality multi-exposure fusion.

2.1.2. Multi-focus image fusion

The problem of multi-focus image fusion stems from the limitations of optical lenses. Concretely, it is difficult to have all objects of different depth-of-field to be all-in-focus within one image. Multi-focus image fusion is to combine images with different focus regions to produce a full-clear image. The CNN and the GAN methods are two main deep learning methods for multi-focus image fusion. Because multi-focus image fusion can be regarded as the selection of clear pixels, these deep learning methods can be divided into the decision map-based method and overall reconstruction-based method.

In essence, the CNN method based on the decision map is a classifier that can effectively determine clarity and blur of pixels, so as to generate the fusion decision map according to the focus condition of source images. The final fused result is obtained by selecting and combining pixels of source images according to the fusion decision map. Strategies for constructing clear and blur image pairs used to train the classifier are important for the final fusion performance. Most methods in this category use the Gaussian kernel to blur the clear image to generate training image pairs [45, 46, 47, 48, 49, 50]. However, this simple Gaussian blur strategy cannot simulate the defocus spread effect of real data. After realizing this problem, some methods [51, 52] use the matte boundary defocus mode to generate realistic training data, and bring better fusion performance. Some other methods [53] consider it unnecessary to construct training image pairs.

165 Instead, they use some simple strategies, such as the guided filtering [54] and repeated blur, to
166 construct rough pseudo-reference decision graphs, guiding the optimization of classification net-
167 works. Then some extra regular constraints are used to refine the decision map [55]. Unlike the
168 above methods, the CNN method based on the overall reconstruction requires the fused image to
169 approach the source image in the corresponding position or region through the screening rules, so
170 as to guide the network to reconstruct the full-clear result [56].

171 The GAN method [57] based on the decision map usually uses the decision map produced by
172 the generator to obtain the fused result, and then uses adversarial learning to make the fused result
173 close to the reference full-clear image. This is essentially a further refinement of the decision map
174 obtained by the CNN method at the image level. The GAN method based on the overall recon-
175 struction is dedicated to forcing the fused image to approach the desired probability distribution at
176 the pixel or feature level, which can further optimize the reconstructed fused result to have richer
177 textures and higher pixel fidelity [58].

178 In general, the decision map-based method and the overall reconstruction-based method have
179 their own advantages and disadvantages. First of all, because the decision map-based method is to
180 directly combine the regions of source images, it can maintain high pixel fidelity. However, such
181 an operation requires very high accuracy of the decision map, which also causes many decision
182 map-based methods to lose scene information near the boundary between focus and non-focus.
183 Moreover, most of these methods require post-processing to refine the generated decision map,
184 such as consistency verification, fully connected conditional random fields (CRF), *etc.* On the
185 contrary, the overall reconstruction-based method has a strong ability to preserve scene informa-
186 tion, but may suffer from a certain degree of pixel value distortion.

187 2.2. *Multi-modal image fusion*

188 The imaging principles of different sensors are diverse, and the focus in describing scene of the
189 multi-modal images captured by them is significantly different. By fusing the complementary and
190 beneficial information in different modal images, it can provide a more comprehensive description
191 of the imaging scene or target. Two most representative multi-modal image tasks include infrared
192 and visible image fusion, and medical image fusion.

2.2.1. Infrared and visible image fusion

The infrared image has significant contrast, which can effectively highlight the target from the background even in bad weather. The visible image contains rich texture details, which are more in line with human visual perception. Infrared and visible image fusion is to combine these two characteristics to produce results with high contrast and rich textures. In order to achieve this goal, the AE, CNN and GAN methods are all introduced into this task.

The AE methods first train an autoencoder on a public dataset such as MS-COCO, where the encoder is dedicated to extracting effective features from the input image, while the decoder is to reconstruct the input image from the encoded features [59, 60]. Then it is natural that the trained autoencoder can be used to solve two sub-problems in image fusion: feature extraction and image reconstruction. Therefore, the key to the image fusion lies in the design of feature fusion strategies. At present, in infrared and visible image fusion, the strategies for feature fusion are still hand-calculated, which are not learnable, such as addition, l_1 -norm [19], attention weighting [18]. This hand-calculated fusion strategy is rough, which limits the further improvement of the infrared and visible image fusion. One type of the CNN methods for infrared and visible image fusion is to realize the three sub-problems end-to-end. For this technical route, the loss function and network structure have a great influence on the final fusion performance [61, 62, 63]. For the design of network structure, the most common and effective means are the residual connection, the dense connection and the dual-stream architecture. Because there is no ground truth in the infrared and visible image fusion, the design of the loss function lies in characterizing the similarity of the contrast and the texture between the fused result and source images. The mainstream loss function terms include the intensity loss, gradient loss, SSIM loss and perceptual loss, and their weight ratio determines the trend of information fusion. Another form of CNN participating in the infrared and visible image fusion is to use a pre-trained network such as VGGNet to extract features from source images, and generate the fusion weight map based on these features [64]. From this perspective, the convolution neural network only realizes the fusion, regardless of feature extraction and image reconstruction, bringing very limited fusion performance.

The GAN methods are currently the most popular in the infrared and visible image fusion, which are able to fulfill feature extraction, feature fusion and image reconstruction in an implicit

manner. Generally, the GAN methods rely on two types of loss functions, namely content loss and adversarial loss. The content loss is similar to that in the CNN methods, which is used to initially fuse source images, while the adversarial loss further constrains the tendency of information fusion. The early GAN methods [24, 65] used for the infrared and visible image fusion only establish the adversarial game between the fused image and the visible image to further enhance the preservation of the abundant details of the visible image. In order to better balance infrared and visible information, subsequent methods [25, 66, 67, 68, 69] have begun to use a single discriminator with multiple classification constraints or dual discriminators to simultaneously estimate the two probability distributions of source images. In general, the GAN methods can produce promising fused results. However, it is not easy to maintain the balance between the generator and the discriminator during training.

2.2.2. Medical image fusion

Medical images can be divided into structural images and functional images according to the information represented. For example, the PET and SPECT images can describe the intensity of body's metabolic functions, while the MRI and CT images can reflect the structure of tissues. The green fluorescent protein (GFP) image can display the functional information related to the molecular distribution of biological living cells, while the phase contrast (PC) image can describe the structural information such as nucleus and mitochondria. Medical image fusion combines two different types of medical images to generate a single image with richer information, which is conducive to more accurate diagnosis of diseases. Currently, two types of popular methods used to achieve medical image fusion are the CNN and the GAN methods.

Most of the CNN methods used for medical image fusion only participate in feature fusion [22, 23, 70]. Specifically, these methods usually use pre-trained convolutional neural networks to measure the activity level of medical image pixels and generate the fusion weight map. Then, the generated weight map is combined with the traditional decomposition and reconstruction strategy, such as Laplacian pyramid, to achieve the medical image fusion. A question that cannot be ignored is that the fusion weight map generated by the pre-trained network may not be appropriate, since the parameters of the network are not trained on medical images. In contrast,

some other CNN methods construct loss functions between the fused result and source images based on the desired information, training the network elaborately on medical images [71, 72]. The trained network can complete feature extraction, feature fusion and image reconstruction end-to-end, which can achieve relatively better fusion performance compared to the above-mentioned type of CNN methods.

The GAN methods simultaneously model the salient information in medical source images through the adversarial learning mechanism [73]. For example, the salient information of functional medical images is the intensity distribution, and the salient information of structural medical images is the spatial texture. GFPPC-GAN [74] introduces GAN into the fusion of GFP and PC images, which uses the adversarial learning between the PC image and the fused image to strengthen the preservation of structural information. DDcGAN [25] adopts multiple discriminators to establish an adversarial game with the generator, so that the fused image produced by the generator can contain function and structure information. Based on DDcGAN, MGMDcGAN [75] introduces a second adversarial game according to a mask calculated to further enhance the preservation of bone dense information. Overall, the GAN methods can achieve excellent performance in medical image fusion. However, because the pixel intensity of the functional image is much larger than the structural image, the texture information is likely to be masked by the function information. This characteristic of medical image fusion poses a challenge to the GAN methods, as the training imbalance of GAN often occurs.

2.3. Sharpening fusion

Sharpening fusion is an effective technology to overcome the contradiction between the spectral and spatial resolution. Under the guidance of spatial images, it is committed to implementing the super-resolution in the spatial dimension while maintaining the spectral resolution. Two typical sharpening fusion tasks are multi-spectral image sharpening and hyperspectral image sharpening.

2.3.1. Multi-spectral sharpening

Multi-spectral sharpening is to fuse the multi-spectral image with low spatial resolution (LRMS) and the panchromatic (PAN) image to produce the multi-spectral image with high spatial resolution (HRMS). Compared with the above mentioned image fusion tasks, the multi-spectral sharpen-

ing requires higher information fidelity (*i.e.*, spectrum, structure). The deep learning methods for sharpening are mainly the CNN and the GAN methods, which are divided into supervised models and unsupervised models according to the way of learning. Specifically, the supervised methods follow the Wald’s protocol [76] to perform spatial degradation operations on the original MS and PAN images to obtain the LRMS and LRPAN images, such as low-pass filter, down-sampling, *etc.* Then the generated LRMS and LRPAN images are used as the input images, and the original MS image is regarded as the reference image. In contrast, the unsupervised methods are directly trained on the original MS and PAN images.

The distance of the pixel intensity between the generated HRMS image and the reference image is the main constraint of the supervised CNN training network. However, the fused results obtained only under this constraint often suffer the spatial structure loss [77, 78, 79]. Although the ℓ_1 distance can protect these texture details better than the ℓ_2 distance, it is still not satisfactory [80, 81]. To address this challenge, some methods adopt novel learning strategies (*e.g.*, the residual learning [82], training in high-frequency domain) and additional loss functions (*e.g.*, the perceptual loss) to further enhance the preservation of structures [83, 84, 85]. Some other methods advocate exploring the spatial structure relationship between the PAN and HRMS images, and use it to guide the fused results to contain more reasonable texture details [86, 87]. The unsupervised CNN methods rely on the loss construction that preserves the spectrum and spatial structures. The usual loss function for spectral preservation is the distance between the fused image after spatial degradation and the LRMS image. Thus the design of loss function for preservation of spatial structures is the key to realizing high-quality sharpening. Generally, the spatial loss is defined as the distance of intensity/gradient between the PAN image and fused HRMS image after transforming, in which the transformation strategies include the regression linear weighting [88], channel replica [16], *etc.*

Compared with the CNN methods, the GAN methods take the adversarial mechanism as an additional constraint to make the fused HRMS image have higher spectral fidelity and reasonable spatial structures. The GAN methods can also be divided into two types: the supervised and unsupervised. The typical supervised GAN method directly uses the reference image as the adversarial target [89, 90], so as to fulfill the estimation of the desired spectral distribution and spatial

structure distributions simultaneously. Because there is no reference target that can provide the ideal spectral and spatial distribution at the same time, the unsupervised GAN method uses two separate discriminators to achieve the preservation of the spectrum and structure [91], in which the data construction involved in the adversarial game is similar to that in the above unsupervised CNN method.

In terms of fusion performance, the current supervised method is better than the unsupervised method. However, it should be noted that the performance of the supervised fusion network is highly dependent on the construction of paired data, which means that the supervised model may not perform well in the full-resolution testing. For the unsupervised method, the main problem at present is that the assumptions used to construct the spatial structure loss are not reasonable enough, which limits the performance of the unsupervised models.

2.3.2. *Hyperspectral sharpening*

Compared with the multi-spectral image, the hyperspectral image has higher spectral resolution and lower spatial resolution. Therefore, hyperspectral sharpening can be realized by fusing the hyperspectral image with low spatial resolution (LRHS) and the multi-spectral image or panchromatic image, so as to produce the hyperspectral image with high spatial resolution (HRHS). The deep models for hyperspectral sharpening include the CNN and the GAN methods. Similar to the multi-spectral sharpening task, these methods in hyperspectral sharpening task can also be divided into supervised and unsupervised approaches.

Since HRHS images are not available, the supervised CNN methods are generally trained on the simulated data. More specifically, these methods directly use the existing spectral response function (SRF) and custom point spread function (PSF) to implement spectral degradation and spatial degradation on the original HS image. The degraded images and the original HS images are constructed into image pairs to learn hyperspectral sharpening under supervision [92, 93, 94, 95]. However, such a model obtained through supervised training in simulation may not be suitable for real scenarios. In some unsupervised CNN methods, although the reference image are not needed to supervise the training process, the CRF and PSF are required to be known. These methods establish the consistency relationship between the generated HRHS image and the observed source

images through the SRF and PSF [96, 97, 98], so as to realize effective estimation of the HRHS image. Unfortunately, the SRF and PSF are unknowable in most cases. In order to address this challenge, some CNN methods combine the hyperspectral unmixing to design coupling networks to estimate the SRF and PSF [99, 100, 101], so as to fulfill the hyperspectral sharpening without supervisor, which have shown promising performance. Different from the above methods, [98] assumes that HRHS and LRHS images share the same spectral subspace, in which the CNN is only used to regularize the estimation of the subspace coefficients, so as to generate the desired HRHS image.

Compared with the CNN method, there is relatively little research on the GAN method for the hyperspectral sharpening. HPGAN [102] adopts the 3D convolutional network to focus on capturing desirable high-frequency residuals under the supervision of reference images. The final fused result is obtained by adding up-sampled LRHS image and the learned high-frequency residual. In addition, HPGAN uses the known SRF to degenerate the generated HRHS image into a panchromatic image, thereby establishing an adversarial game with the source PAN image, which further enhances the performance of hyperspectral sharpening.

2.4. Unified image fusion approaches

Some methods have good universality, which can achieve various image fusion tasks using a unified framework. Although the ideas of these methods are often very different, their characteristic is that they can find the commonality of different fusion tasks. In particular, U2Fusion [1] adopts the elastic weight consolidation (EWC) to continuously train the fusion network, which can overcome the catastrophic forgetting problem in cross-task learning. In this way, U2Fusion establishes a general model for various image fusion tasks. PMGI [16] models a variety of image fusion tasks as the extraction and reconstruction of intensity and gradient information. It is only necessary to adjust the coefficient ratio of the loss function items according to the tendency of information fusion in the specific task to adapt to multiple image fusion scenarios. IFCNN [17] believes any image fusion task is a problem of effective information selection, in which the evaluation of information effectiveness is based on the sharpness of pixels. Similar unified approaches include SGRFR [103], CU-Net [104], DIF-Net [105], *etc.*

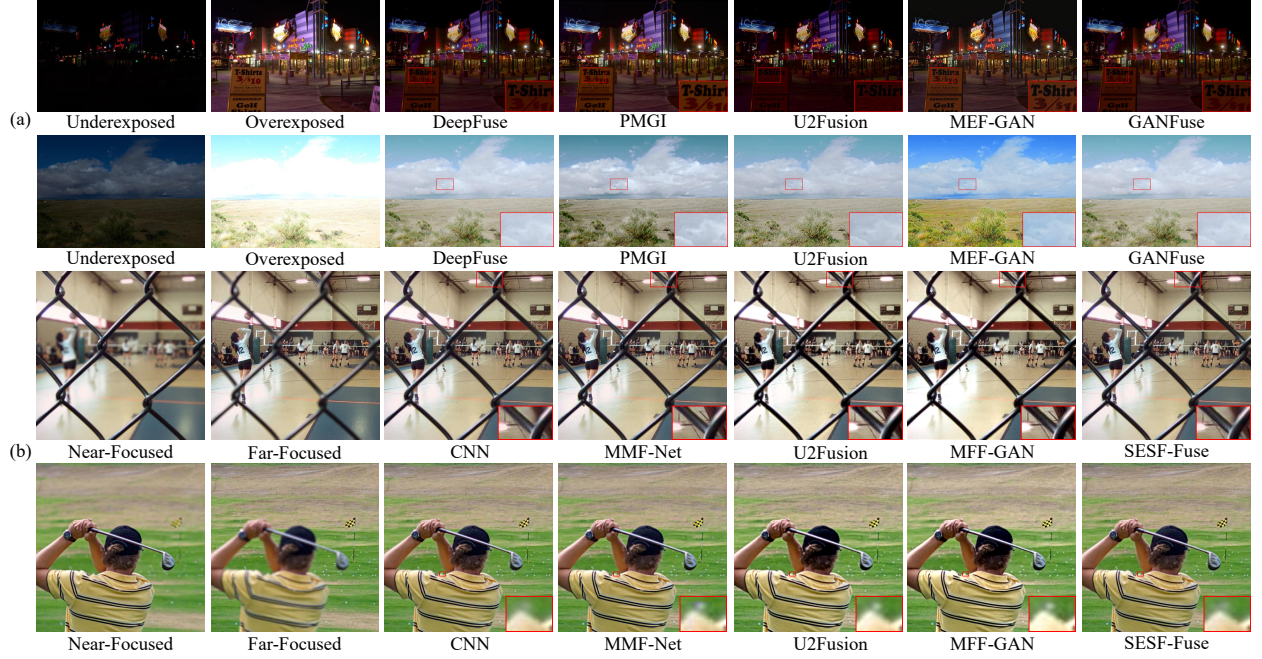


Figure 4: Qualitative comparisons of representative methods in digital photography image fusion scenarios. (a) and (b) are the multi-exposure image fusion and the multi-focus image fusion, respectively.

3. Evaluation

For six image fusion tasks mentioned above, we conduct evaluation experiments on their respective representative datasets. Specifically, the testing data in multi-exposure image fusion are from *MEF*¹ dataset [107], which collects multi-exposure sequences from two categories of scenes (indoor and outdoor); the testing data in multi-focus image fusion are from *Lytro*² dataset [108], which uses the Lytro camera to change the depth of focus under specific viewing angles to obtain digital images with different focus regions; the testing data in infrared and visible image fusion are from *TNO*³ dataset, which mainly describes various military related scenes; the testing data in medical image fusion are from *Harvard*⁴ medical school website, which includes different transaxial sections of the brain-hemispheric; the testing data in multi-spectral sharpening are from

¹<https://github.com/csajcai/SICE>

²<https://mansournejati.ece.iut.ac.ir/content/lytro-multi-focus-dataset>

³https://figshare.com/articles/TNO_Image_Fusion_Dataset/1008029

⁴<https://www.med.harvard.edu/AANLIB/home.html>

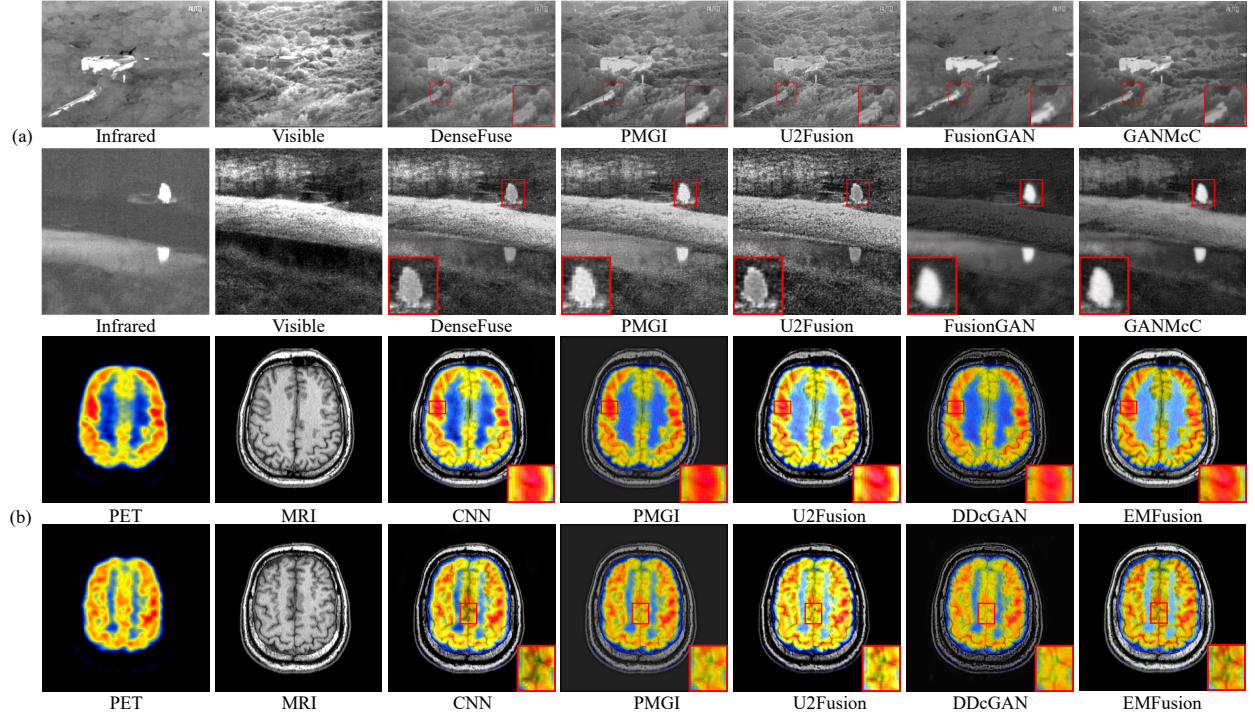


Figure 5: Qualitative comparisons of representative methods in multi-modal image fusion scenarios. (a) and (b) are the infrared and visible image fusion and the medical image fusion, respectively.

QuickBird dataset, in which the multi-spectral image contains 4 bands; the testing data in hyper-spectral sharpening are from *Cave*⁵ dataset [109], in which the hyperspectral image contains 31 bands. Five representative methods in each image fusion task are selected for evaluation, including AE-based methods, CNN-based methods, GAN-based methods and unified methods.

3.1. Qualitative and quantitative evaluation

The evaluation is firstly performed from both qualitative and quantitative aspects. For qualitative evaluation, two typical visualization results are selected for each image fusion task, in which relevant regions are highlighted to reflect the difference. For quantitative evaluation, we select 4 popular objective metrics for each fusion scenario according to their characteristics. The qualitative and quantitative results are reported in Figs. 4-6 and Table 1.

⁵<https://www.cs.columbia.edu/CAVE/databases/multispectral>

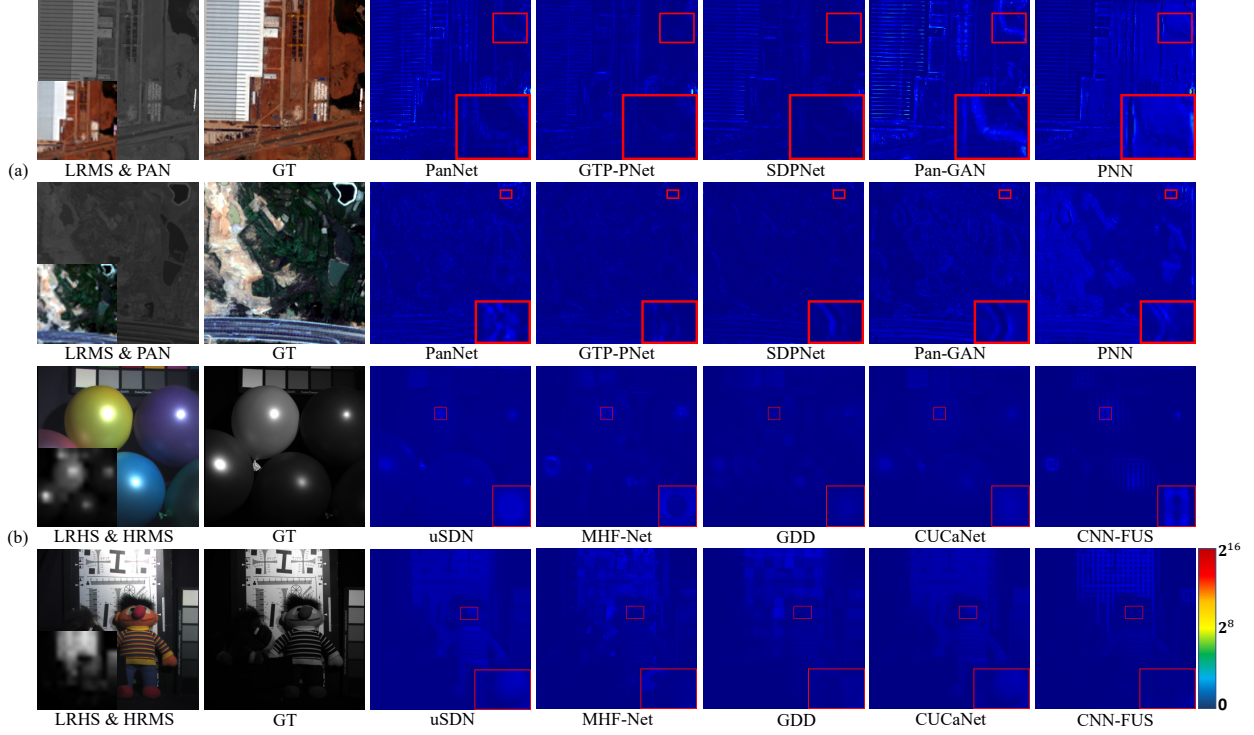


Figure 6: Qualitative comparisons of representative methods in sharpening fusion scenarios. (a) and (b) are the multi-spectral sharpening and the hyperspectral sharpening, respectively. We show the residuals between the fused result and the ground truth.

In digital photography image fusion, DeepFuse [21], GANFuse [44], CNN [45] and MMF-Net [51] achieve relatively good performance, which derives two routes to further improve the performance of deep learning. One is to design higher-quality non-reference metrics for unsupervised learning, and the other is to construct data more in line with the real situation for supervised training. It can be found that some unified approaches such as PMGI [16] and U2Fusion [1] are even better than specially designed methods in some multi-modal image fusion tasks. This phenomenon shows that it is very promising to seek the commonality of image fusion scenarios to establish a general model. In sharpening fusion scenarios, the common characteristic of methods that achieve better performance is that they fully consider the relationship between the fused result and source images in the spatial or spectral dimension, such as GTP-PNet [87] and GDD [96]. This phenomenon encourages subsequent research to pay more attention to the estimation of spatial and spectral degradation models.

Table 1: Quantitative comparisons of representative methods in six fusion tasks. We select different metrics according to the characteristics of different fusion scenarios to achieve targeted evaluation. **Bold** indicates the best result.

Digital	Multi-exposure (MEF)					Multi-focus (Lytro)				
Photography	DeepFuse [21]	PMGI [16]	U2Fusion [1]	MEF-GAN [43]	GANFuse [44]	CNN [45]	MMF-Net [51]	U2Fusion [1]	MFF-GAN [58]	SESF-Fuse [53]
$Q^{AB/F} \uparrow$	0.633±0.043	0.628±0.058	0.612±0.058	0.544±0.059	0.628±0.048	0.535±0.083	0.697±0.039	0.655±0.030	0.662±0.025	0.526±0.085
SSIM $\uparrow$	0.665±0.060	0.654±0.060	0.637±0.060	0.634±0.067	0.665±0.061	0.844±0.062	0.844±0.060	0.815±0.058	0.829±0.055	0.842±0.062
PSNR $\uparrow$	10.510±1.576	10.393±1.490	10.436±1.382	10.472±1.523	10.512±1.585	27.133±3.088	26.884±2.922	24.578±1.434	24.929±1.827	27.017±3.091
$N^{AB/F} \downarrow$	0.090±0.039	0.150±0.103	0.237±0.056	0.266±0.073	0.092±0.039	0.028±0.035	0.118±0.030	0.174±0.024	0.169±0.026	0.030±0.036
Multi-Modal	Infrared-Visible (TNO)					Medical (Harvard)				
	DenseFuse [19]	PMGI [16]	U2Fusion [1]	FusionGAN [24]	GANMcC [67]	CNN [23]	PMGI [16]	U2Fusion [1]	DDcGAN [25]	MEFusion [106]
EN $\uparrow$	6.826±0.374	7.031±0.299	6.956±0.336	6.546±0.409	6.698±0.405	4.862±0.195	5.985±0.234	5.416±0.222	5.374±0.213	5.106±0.231
SSIM $\uparrow$	0.722±0.055	0.710±0.054	0.676±0.058	0.656±0.061	0.688±0.065	0.509±0.004	0.411±0.012	0.530±0.005	0.492±0.003	0.545±0.004
PSNR $\uparrow$	14.786±2.722	14.432±2.304	14.984±2.679	13.000±2.007	13.696±2.579	8.163±0.153	8.179±0.154	8.170±0.153	8.165±0.153	8.168±0.153
SF $\uparrow$	0.034±0.012	0.033±0.012	0.044±0.014	0.023±0.007	0.023±0.007	0.049±0.002	0.118±0.005	0.093±0.005	0.075±0.002	0.095±0.005
Sharpening	Multi-spectral Sharpening (QuickBird)					Hyperspectral Sharpening (Cave)				
	PanNet [83]	GTP-PNet [87]	SDPNet [84]	Pan-GAN [91]	PNN [77]	uSDN [97]	MHF-Net [93]	GDD [96]	CUCaNet [99]	CNN-FUS [98]
RMSE $\downarrow$	4.380±0.658	3.174±0.504	3.395±0.595	4.988±0.836	6.442±0.650	5.815±0.402	4.755±1.716	3.773±1.706	4.564±1.469	4.721±2.108
SAM $\downarrow$	3.291±1.085	1.938±0.435	2.561±0.928	3.606±1.993	4.296±0.697	11.080±3.451	6.435±2.139	5.630±2.845	8.245±1.530	6.526±3.437
PSNR $\uparrow$	36.051±1.574	38.993±1.591	38.109±1.792	34.785±1.504	32.159±0.833	34.932±3.023	35.112±3.084	37.157±4.114	35.671±2.977	36.412±3.816
ERGAS $\downarrow$	1.926±0.482	1.385±0.341	1.512±0.415	2.194±0.482	3.059±0.888	0.486±0.027	0.380±0.113	0.304±0.124	0.389±0.120	0.394±0.164

3.2. Efficiency evaluation

For deep learning-based fusion methods, running efficiency is an important factor in measuring fusion performance. We count the running time of the typical methods above, and the results are reported in Table 2. It can be seen that the unified PMGI [16], GAN-based MFF-GAN [58], CNN-based PNN [77] and CuCaNet [99] achieve relatively high running efficiency, which is due to their common characteristics. Specifically, these methods all have relatively simple network structures, which ensure a low amount of model parameters. Under simple network structures, designing constraints and optimization approaches that are more suitable for task characteristics can obtain a double harvest of accuracy and efficiency. This enlightens that subsequent researches should pay more attention to the characteristics of the image fusion task itself, rather than blindly increase the scale of neural networks.

Table 2: Average running time of representative methods in six fusion tasks (unit: second). **Bold** indicates the best result.

Digital Photography	Multi-Exposure (MEF) (Mean Size: 600×1000)	DeepFuse [21]	PMGI [16]	U2Fusion [1]	MEF-GAN [43]	GANFuse [44]
		0.227±0.033	0.204±0.117	1.579±1.513	3.673±2.522	0.447±0.305
	Multi-Focus (Lytro) (Mean Size: 520×520)	CNN [45]	MMF-Net [51]	U2Fusion [1]	MFF-GAN [58]	SESF-Fuse [53]
		120.315±6.291	1.237±0.744	0.503±0.623	0.284±0.690	0.366±0.023
Multi-Modal	Infrared-Visible (TNO) (Mean Size: 576×768)	DenseFuse [19]	PMGI [16]	U2Fusion [1]	FusionGAN [24]	GANMcC [67]
		0.321±0.122	0.099±0.076	0.636±0.536	0.170±0.129	0.127±0.059
	Medical (Harvard) (Mean Size: 256×256)	CNN [23]	PMGI [16]	U2Fusion [1]	DDcGAN [25]	MEFusion [106]
		16.504±0.562	0.025±0.011	0.206±0.035	0.948±0.017	0.060±0.059
Sharpening	MS Sharpening (QuickBird) (Mean Size: 200×200)	PanNet [83]	GTP-PNet [87]	SDPNet [84]	Pan-GAN [91]	PNN [77]
		0.100±0.189	0.010±0.013	0.018±0.072	0.042±0.021	0.009±0.047
	HS Sharpening (Cave) (Mean Size: 512×512)	uSDN [97]	MHF-Net [93]	GDD [96]	CUCaNet [99]	CNN-FUS [98]
		209.249±104.165	2.680±3.493	2608.036±333.362	0.070±0.001	216.559±1.567

4. Applications

Image fusion can effectively integrate information from different source images to provide more useful input for downstream tasks, so as to improve the performance of these applications. We will briefly introduce several typical applications, *e.g.*, photography visualization, object tracking, medical diagnosis and remote sensing monitoring, which can intuitively demonstrate the importance of image fusion.

4.1. Photography visualization

Photography visualization is to display the captured results of digital imaging devices well, being dedicated to improving the visual experience of users [110, 111]. However, digital imaging devices often have a predefined depth-of-field and limited dynamic range, which means that the image directly output by the device may not be desired [112, 113]. Image fusion can combine the effective information in the images captured under different shot settings to generate a full-focused image with appropriate exposure, which greatly improves the quality of photography [114, 115], as shown in Fig. 7. At present, image fusion technology has been integrated into some digital



Figure 7: Two typical cases of photography visualization. Image fusion can provide high-quality images with good visual experience.

imaging devices [116, 117], including cameras, mobile phones, *etc.*

4.2. RGBT object tracking

The object tracking is to find the object of interest defined in the current frame in a subsequent frame of a video [118]. The most common one is the single-mode tracking, such as the visible modal-based tracking and infrared modal-based tracking. However, due to the limitation of the single mode representation, these tracking methods are not robust. Specifically, the quality of visible images depends on the imaging environment, which means that the performance of target tracking is not guaranteed at night or in poor lighting conditions. Similarly, infrared images lack textures and have a poor scene stereoscopic sense, that are also unreliable in certain situations. As a result, a new object tracking technology route is derived, saying the RGBT tracking [119, 120], which fuses the complementary information in infrared and visible images to make the object tracking more robust, as shown in Fig. 8. Common RGBT tracking methods include sparse representation-based



Figure 8: A typical case of RGBT object tracking. Image fusion can make the tracked object more distinguishable from the background and other targets.

methods [121, 122], graph-based methods [123, 124], correlation filter-based methods [125, 126], and deep learning-based methods [127, 128]. Regardless of the type of method, maintaining effective target information in the fusion process is the key to achieve high quality RGBT tracking. In other words, the fused result that can retain the target salience in the infrared modal and the target texture details in the visible modal at the same time will make the tracking accuracy more promising [129, 130, 131].

4.3. Medical diagnosis

The vast majority of medical diagnoses are made by computers or doctors analyzing medical images. The imaging mechanisms of medical images in different modalities are diverse, and their emphasis on describing the body information is also different [26, 34]. Common medical modalities include the computerized tomography (CT) [132], magnetic resonance imaging (MRI) [133], positron emission tomography (PET) [134], single photon emission computed tomography (SPECT) [135] and ultrasound [136]. Among them, some modalities focus on describing the structure of organs and tissues, while others focus on describing the intensity of regional metabolism. In this context, fusing medical images of different modalities will greatly improve the accuracy and efficiency of diagnosis while reducing redundant information and improving the image quality [137, 138], as shown in Fig. 9. Specifically, some diseases not only cause changes in the physical shape of tissues, but also increase the intensity of local metabolism, such as tumors. Thus, the combination of functional and structural medical images can improve the diagnosis ac-

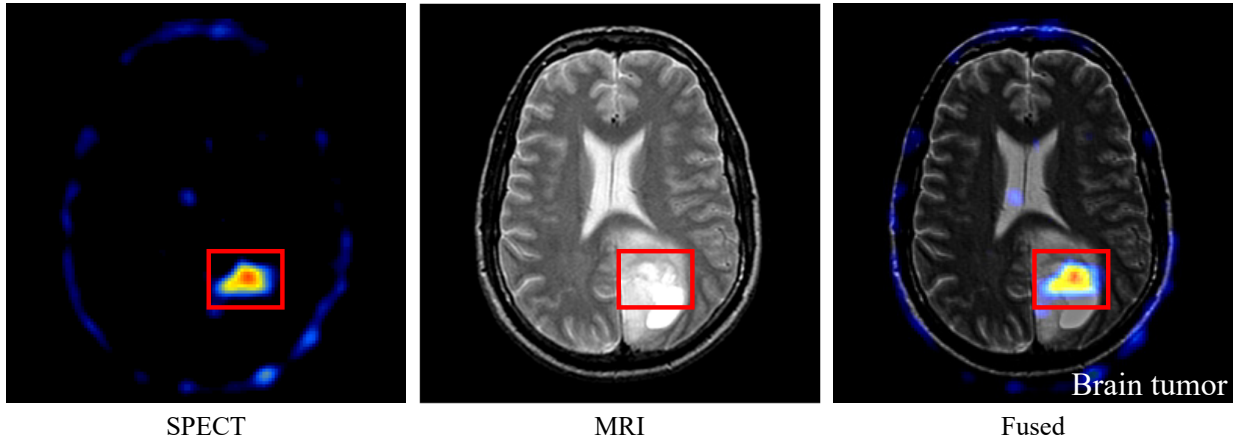


Figure 9: A typical case of medical diagnosis. Image fusion can make the diagnosis more accurate and efficient.

curacy of these diseases [139, 140, 141]. In addition, medical image fusion can promote the synchronous realization of disease determination and lesion localization, which will greatly improve the efficiency of diagnosis and save the time for follow-up treatments. It is precisely because of these attractive benefits that image fusion has been integrated into some medical diagnostic devices to help frontier medical workers fulfill high-quality diagnosis [142].

4.4. Remote sensing monitoring

Remote sensing monitoring refers to the observation and investigation of land, sea and weather by analyzing hyperspectral/multi-spectral images [143, 144]. Common remote sensing monitoring tasks include the land cover classification [145], vegetation detection [146], mineral exploration [147], *etc.* However, the spectral resolution and spatial resolution cannot be achieved at the same time. In other words, the low spatial resolution of hyperspectral/multi-spectral images restricts the further improvement of the accuracy of remote sensing monitoring. Fortunately, sharpening fusion can increase the spatial resolution while maintaining the spectral resolution [148]. Thus, technicians have introduced image fusion to pre-process the remote sensing images, greatly improving the performance of related monitoring tasks, as shown in Fig. 10. Nowadays, sharpening fusion technology has been successfully applied to tasks such as digital earth modeling and land cover mapping, showing good practical value [148, 149].

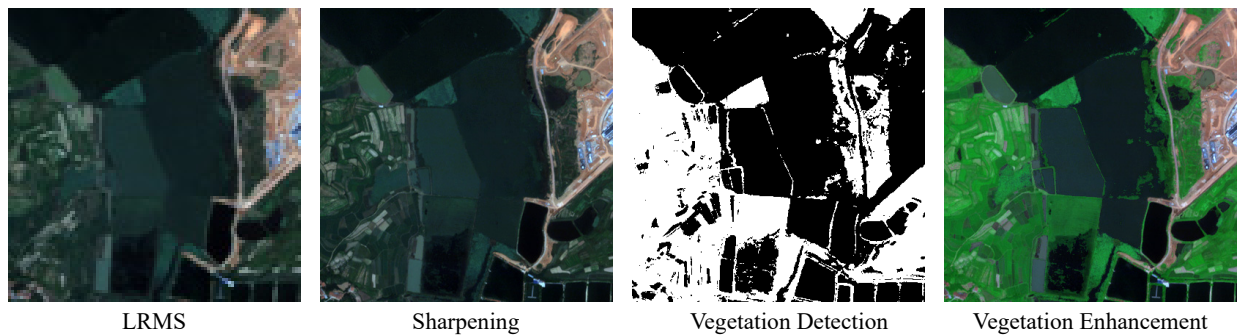


Figure 10: A typical case of remote sensing monitoring. Image fusion can improve the accuracy of the vegetation detection.

5. Conclusion and prospect

As an enhancement technology, image fusion has played a critical role in various fields, including the photography visualization, object tracking, medical diagnosis and remote sensing monitoring. In recent years, an increasing number of image fusion method based on deep learning have been proposed to improve the fusion performance. In order to provide effective reference and understanding for relevant researchers and technicians, this survey comprehensively analyzes the latest deep learning-based methods in the field of image fusion. We discuss the characteristics of various methods in combination with specific image fusion scenarios, in which the directions for further improvement are also provided. In addition, we also conduct performance comparisons between representative approaches in each fusion task, and introduce some typical applications of image fusion. Despite of extensive efforts that have been made for image fusion with deep learning, there are still several open challenges.

Non-registered image fusion. Currently, in almost all image fusion tasks, deep learning-based methods assume that source images are pre-registered [150]. However, both multi-modal images and digital photography images are not registered because of parallax, scale differences and other factors in real scenes. As a result, the operations along the spatial pixel position in the existing deep learning methods are not available for realistic source images. Although many off-the-shelf methods [151] can be used to pre-register the source images, the reliance on the pre-processing of the registration algorithm may lead to certain limitations, such as low efficiency and dependence

on registration accuracy. Therefore, it is desirable to develop the non-registered fusion algorithm to realize image registration and fusion in an implicit manner.

Different resolution image fusion. Generally, the resolution of source images is different due to the difference in principles of sensors. It is a challenge to overcome the resolution difference and make full use of the information in different source images to achieve effective fusion. Although some methods [25, 152] have been proposed to solve different resolution image fusion, there are still some open problems, such as the adopted upsampling strategy and the upsampling location in networks. More importantly, it is desired to design deep networks by organically combining the characteristics of super-resolution and image fusion tasks.

Task-oriented image fusion. The original intention of image fusion is to provide more beneficial input for subsequent applications. However, in many image fusion tasks, the existing deep learning-based methods do not consider the correlation between fusion and subsequent applications when designing the loss function, which often leads to very subjective fused results. Future researches could consider introducing the accuracy of subsequent tasks into the loss function design in the fusion stage to guide the fusion process from the decision-making level.

Real-time image fusion. From the perspective of application requirements, image fusion is the forerunner of many visual tasks, and its performance directly affects the accuracy of entire tasks. Some practical applications have high requirements on the real-time performance of the algorithm. However, the complex transformation decomposition and iterative optimization in current image fusion methods lead to lower operating efficiency, which limits the application of image fusion in some real-time tasks, such as video surveillance. Therefore, developing the real-time image fusion algorithms is of great significance, which will make image fusion have broader application prospects.

Fusion quality assessment. Because there is no real ground truth in most image fusion tasks, it is very challenging to evaluate the quality of the fused result. Therefore, it is very important for the image fusion field to design no-reference metrics with more characterization ability. On the one hand, the proposed metrics can be used to construct the loss function to guide higher-quality fusion. On the other hand, the newly designed metrics can also fairly evaluate the fused results to encourage subsequent fusion research. The distance metric learning may be a good choice for

fusion quality assessment.

Based on the above review and prospect, we have not reached the upper bound of image fusion. The non-registered fusion method is more robust to real data; the different resolution image fusion can overcome the scale differences; the task-oriented fusion method has more practical value; the real-time image fusion has broader application prospects; the higher-quality fusion assessment metrics are able to improve fusion performance. Let's wait and see.

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References

- [1] H. Xu, J. Ma, J. Jiang, X. Guo, H. Ling, U2fusion: A unified unsupervised image fusion network, *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2020).
- [2] J. Ma, Y. Ma, C. Li, Infrared and visible image fusion methods and applications: A survey, *Information Fusion* 45 (2019) 153–178.
- [3] G. Petschnigg, R. Szeliski, M. Agrawala, M. Cohen, H. Hoppe, K. Toyama, Digital photography with flash and no-flash image pairs, *ACM Transactions on Graphics* 23 (2004) 664–672.
- [4] Y. Zhu, C. Li, B. Luo, J. Tang, X. Wang, Dense feature aggregation and pruning for rgbt tracking, in: *Proceedings of the ACM International Conference on Multimedia*, 2019, pp. 465–472.
- [5] G. Bhatnagar, Q. J. Wu, Z. Liu, Directive contrast based multimodal medical image fusion in nsct domain, *IEEE Transactions on Multimedia* 15 (2013) 1014–1024.
- [6] D. Amarsaikhan, M. Saandar, M. Ganzorig, H. Blotevogel, E. Egshiglen, R. Gantuyal, B. Nergui, D. Enkhjargal, Comparison of multisource image fusion methods and land cover classification, *International Journal of Remote Sensing* 33 (2012) 2532–2550.
- [7] G. Pajares, J. M. De La Cruz, A wavelet-based image fusion tutorial, *Pattern Recognition* 37 (2004) 1855–1872.
- [8] S. Li, B. Yang, J. Hu, Performance comparison of different multi-resolution transforms for image fusion, *Information Fusion* 12 (2011) 74–84.
- [9] Y. Mo, X. Kang, P. Duan, B. Sun, S. Li, Attribute filter based infrared and visible image fusion, *Information Fusion* 75 (2021) 41–54.

- [10] S. Li, X. Kang, J. Hu, Image fusion with guided filtering, *IEEE Transactions on Image Processing* 22 (2013) 2864–2875.
- [11] Y. Liu, S. Liu, Z. Wang, A general framework for image fusion based on multi-scale transform and sparse representation, *Information Fusion* 24 (2015) 147–164.
- [12] B. Yang, S. Li, Multifocus image fusion and restoration with sparse representation, *IEEE Transactions on Instrumentation and Measurement* 59 (2009) 884–892.
- [13] J. C. Harsanyi, C.-I. Chang, Hyperspectral image classification and dimensionality reduction: An orthogonal subspace projection approach, *IEEE Transactions on Geoscience and Remote Sensing* 32 (1994) 779–785.
- [14] J. Han, E. J. Pauwels, P. De Zeeuw, Fast saliency-aware multi-modality image fusion, *Neurocomputing* 111 (2013) 70–80.
- [15] J. Ma, C. Chen, C. Li, J. Huang, Infrared and visible image fusion via gradient transfer and total variation minimization, *Information Fusion* 31 (2016) 100–109.
- [16] H. Zhang, H. Xu, Y. Xiao, X. Guo, J. Ma, Rethinking the image fusion: A fast unified image fusion network based on proportional maintenance of gradient and intensity, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, 2020, pp. 12797–12804.
- [17] Y. Zhang, Y. Liu, P. Sun, H. Yan, X. Zhao, L. Zhang, Ifcnn: A general image fusion framework based on convolutional neural network, *Information Fusion* 54 (2020) 99–118.
- [18] H. Li, X.-J. Wu, T. Durrani, Nestfuse: An infrared and visible image fusion architecture based on nest connection and spatial/channel attention models, *IEEE Transactions on Instrumentation and Measurement* 69 (2020) 9645–9656.
- [19] H. Li, X.-J. Wu, Densefuse: A fusion approach to infrared and visible images, *IEEE Transactions on Image Processing* 28 (2019) 2614–2623.
- [20] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, C. L. Zitnick, Microsoft coco: Common objects in context, in: *Proceedings of the European Conference on Computer Vision*, 2014, pp. 740–755.
- [21] K. Ram Prabhakar, V. Sai Srikar, R. Venkatesh Babu, Deepfuse: A deep unsupervised approach for exposure fusion with extreme exposure image pairs, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 4714–4722.
- [22] K. Wang, M. Zheng, H. Wei, G. Qi, Y. Li, Multi-modality medical image fusion using convolutional neural network and contrast pyramid, *Sensors* 20 (2020) 2169.
- [23] Y. Liu, X. Chen, J. Cheng, H. Peng, A medical image fusion method based on convolutional neural networks, in: *Proceedings of the International Conference on Information Fusion*, 2017, pp. 1–7.
- [24] J. Ma, W. Yu, P. Liang, C. Li, J. Jiang, Fusiongan: A generative adversarial network for infrared and visible image fusion, *Information Fusion* 48 (2019) 11–26.

- [25] J. Ma, H. Xu, J. Jiang, X. Mei, X.-P. Zhang, Ddcgan: A dual-discriminator conditional generative adversarial network for multi-resolution image fusion, *IEEE Transactions on Image Processing* 29 (2020) 4980–4995.
- [26] A. P. James, B. V. Dasarathy, Medical image fusion: A survey of the state of the art, *Information Fusion* 19 (2014) 4–19.
- [27] X. Zhang, Deep learning-based multi-focus image fusion: A survey and a comparative study, *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2021).
- [28] Q. Zhang, Y. Liu, R. S. Blum, J. Han, D. Tao, Sparse representation based multi-sensor image fusion for multi-focus and multi-modality images: A review, *Information Fusion* 40 (2018) 57–75.
- [29] H. Ghassemian, A review of remote sensing image fusion methods, *Information Fusion* 32 (2016) 75–89.
- [30] S. Bhat, D. Koundal, Multi-focus image fusion techniques: a survey, *Artificial Intelligence Review* (2021) 1–53.
- [31] Y. Liu, L. Wang, J. Cheng, C. Li, X. Chen, Multi-focus image fusion: A survey of the state of the art, *Information Fusion* 64 (2020) 71–91.
- [32] S. Li, X. Kang, L. Fang, J. Hu, H. Yin, Pixel-level image fusion: A survey of the state of the art, *Information Fusion* 33 (2017) 100–112.
- [33] B. Meher, S. Agrawal, R. Panda, A. Abraham, A survey on region based image fusion methods, *Information Fusion* 48 (2019) 119–132.
- [34] J. Du, W. Li, K. Lu, B. Xiao, An overview of multi-modal medical image fusion, *Neurocomputing* 215 (2016) 3–20.
- [35] F. D. Javan, F. Samadzadegan, S. Mehravar, A. Toosi, R. Khatami, A. Stein, A review of image fusion techniques for pan-sharpening of high-resolution satellite imagery, *ISPRS Journal of Photogrammetry and Remote Sensing* 171 (2021) 101–117.
- [36] Y. Liu, X. Chen, Z. Wang, Z. J. Wang, R. K. Ward, X. Wang, Deep learning for pixel-level image fusion: Recent advances and future prospects, *Information Fusion* 42 (2018) 158–173.
- [37] S. K. Nayar, T. Mitsunaga, High dynamic range imaging: Spatially varying pixel exposures, in: *Proceedings IEEE Conference on Computer Vision and Pattern Recognition*, 2000, pp. 472–479.
- [38] H. Li, L. Zhang, Multi-exposure fusion with cnn features, in: *Proceedings of the IEEE International Conference on Image Processing*, 2018, pp. 1723–1727.
- [39] Z. Pan, M. Yu, G. Jiang, H. Xu, Z. Peng, F. Chen, Multi-exposure high dynamic range imaging with informative content enhanced network, *Neurocomputing* 386 (2020) 147–164.
- [40] K. Ma, K. Zeng, Z. Wang, Perceptual quality assessment for multi-exposure image fusion, *IEEE Transactions on Image Processing* 24 (2015) 3345–3356.
- [41] K. Ma, Z. Duanmu, H. Zhu, Y. Fang, Z. Wang, Deep guided learning for fast multi-exposure image fusion, *IEEE Transactions on Image Processing* 29 (2020) 2808–2819.

- [42] Y. Qi, S. Zhou, Z. Zhang, S. Luo, X. Lin, L. Wang, B. Qiang, Deep unsupervised learning based on color un-referenced loss functions for multi-exposure image fusion, *Information Fusion* 66 (2021) 18–39.
- [43] H. Xu, J. Ma, X.-P. Zhang, Mef-gan: Multi-exposure image fusion via generative adversarial networks, *IEEE Transactions on Image Processing* 29 (2020) 7203–7216.
- [44] Z. Yang, Y. Chen, Z. Le, Y. Ma, Ganfuse: a novel multi-exposure image fusion method based on generative adversarial networks, *Neural Computing and Applications* 33 (11) (2021) 6133–6145.
- [45] Y. Liu, X. Chen, H. Peng, Z. Wang, Multi-focus image fusion with a deep convolutional neural network, *Information Fusion* 36 (2017) 191–207.
- [46] J. Li, X. Guo, G. Lu, B. Zhang, Y. Xu, F. Wu, D. Zhang, Drpl: Deep regression pair learning for multi-focus image fusion, *IEEE Transactions on Image Processing* 29 (2020) 4816–4831.
- [47] H. Tang, B. Xiao, W. Li, G. Wang, Pixel convolutional neural network for multi-focus image fusion, *Information Sciences* 433 (2018) 125–141.
- [48] M. Amin-Naji, A. Aghagolzadeh, M. Ezoji, Ensemble of cnn for multi-focus image fusion, *Information Fusion* 51 (2019) 201–214.
- [49] Y. Yang, Z. Nie, S. Huang, P. Lin, J. Wu, Multilevel features convolutional neural network for multifocus image fusion, *IEEE Transactions on Computational Imaging* 5 (2019) 262–273.
- [50] B. Xiao, B. Xu, X. Bi, W. Li, Global-feature encoding u-net (geu-net) for multi-focus image fusion, *IEEE Transactions on Image Processing* 30 (2020) 163–175.
- [51] H. Ma, Q. Liao, J. Zhang, S. Liu, J.-H. Xue, An α -matte boundary defocus model-based cascaded network for multi-focus image fusion, *IEEE Transactions on Image Processing* 29 (2020) 8668–8679.
- [52] S. Xu, L. Ji, Z. Wang, P. Li, K. Sun, C. Zhang, J. Zhang, Towards reducing severe defocus spread effects for multi-focus image fusion via an optimization based strategy, *IEEE Transactions on Computational Imaging* 6 (2020) 1561–1570.
- [53] B. Ma, Y. Zhu, X. Yin, X. Ban, H. Huang, M. Mukeshimana, Sesf-fuse: An unsupervised deep model for multi-focus image fusion, *Neural Computing and Applications* 33 (11) (2021) 5793–5804.
- [54] K. He, J. Sun, X. Tang, Guided image filtering, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 35 (2012) 1397–1409.
- [55] J. Ma, Z. Le, X. Tian, J. Jiang, Smfuse: Multi-focus image fusion via self-supervised mask-optimization, *IEEE Transactions on Computational Imaging* 7 (2021) 309–320.
- [56] X. Yan, S. Z. Gilani, H. Qin, A. Mian, Structural similarity loss for learning to fuse multi-focus images, *Sensors* 20 (2020) 6647.
- [57] X. Guo, R. Nie, J. Cao, D. Zhou, L. Mei, K. He, Fusegan: Learning to fuse multi-focus image via conditional generative adversarial network, *IEEE Transactions on Multimedia* 21 (2019) 1982–1996.
- [58] H. Zhang, Z. Le, Z. Shao, H. Xu, J. Ma, Mff-gan: An unsupervised generative adversarial network with

- adaptive and gradient joint constraints for multi-focus image fusion, *Information Fusion* 66 (2020) 40–53.
- [59] L. Jian, X. Yang, Z. Liu, G. Jeon, M. Gao, D. Chisholm, Sedrfuse: A symmetric encoder–decoder with residual block network for infrared and visible image fusion, *IEEE Transactions on Instrumentation and Measurement* 70 (2021) 5002215.
- [60] Y. Long, H. Jia, Y. Zhong, Y. Jiang, Y. Jia, Rxdnfnuse: A aggregated residual dense network for infrared and visible image fusion, *Information Fusion* 69 (2021) 128–141.
- [61] R. Hou, D. Zhou, R. Nie, D. Liu, L. Xiong, Y. Guo, C. Yu, Vif-net: an unsupervised framework for infrared and visible image fusion, *IEEE Transactions on Computational Imaging* 6 (2020) 640–651.
- [62] J. Ma, L. Tang, M. Xu, H. Zhang, G. Xiao, Stdffusionnet: An infrared and visible image fusion network based on salient target detection, *IEEE Transactions on Instrumentation and Measurement* 70 (2021) 5009513.
- [63] H. Xu, X. Wang, J. Ma, Drf: Disentangled representation for visible and infrared image fusion, *IEEE Transactions on Instrumentation and Measurement* 70 (2021) 5006713.
- [64] H. Li, X.-J. Wu, J. Kittler, Infrared and visible image fusion using a deep learning framework, in: *Proceedings of the International Conference on Pattern Recognition*, 2018, pp. 2705–2710.
- [65] J. Ma, P. Liang, W. Yu, C. Chen, X. Guo, J. Wu, J. Jiang, Infrared and visible image fusion via detail preserving adversarial learning, *Information Fusion* 54 (2020) 85–98.
- [66] J. Li, H. Huo, C. Li, R. Wang, Q. Feng, Attentionfgan: Infrared and visible image fusion using attention-based generative adversarial networks, *IEEE Transactions on Multimedia* 23 (2021) 1383–1396.
- [67] J. Ma, H. Zhang, Z. Shao, P. Liang, H. Xu, Ganmcc: A generative adversarial network with multi-classification constraints for infrared and visible image fusion, *IEEE Transactions on Instrumentation and Measurement* 70 (2021) 5005014.
- [68] J. Li, H. Huo, K. Liu, C. Li, Infrared and visible image fusion using dual discriminators generative adversarial networks with wasserstein distance, *Information Sciences* 529 (2020) 28–41.
- [69] J. Li, H. Huo, C. Li, R. Wang, C. Sui, Z. Liu, Multigrained attention network for infrared and visible image fusion, *IEEE Transactions on Instrumentation and Measurement* 70 (2021) 5002412.
- [70] F. Lahoud, S. Süssstrunk, Zero-learning fast medical image fusion, in: *Proceedings of the International Conference on Information Fusion*, 2019, pp. 1–8.
- [71] H. Xu, J. Ma, Z. Le, J. Jiang, X. Guo, FusionDn: A unified densely connected network for image fusion., in: *Proceedings of the AAAI Conference on Artificial Intelligence*, 2020, pp. 12484–12491.
- [72] W. Tang, Y. Liu, J. Cheng, C. Li, X. Chen, Green fluorescent protein and phase contrast image fusion via detail preserving cross network, *IEEE Transactions on Computational Imaging* 7 (2021) 584–597.
- [73] C. Zhao, T. Wang, B. Lei, Medical image fusion method based on dense block and deep convolutional generative adversarial network, *Neural Computing and Applications* 33 (12) (2021) 6595–6610.
- [74] W. Tang, Y. Liu, C. Zhang, J. Cheng, H. Peng, X. Chen, Green fluorescent protein and phase-contrast image

fusion via generative adversarial networks, *Computational and Mathematical Methods in Medicine* 2019 (2019) 5450373.

- [75] Z. Le, J. Huang, F. Fan, X. Tian, J. Ma, A generative adversarial network for medical image fusion, in: *Proceedings of the IEEE International Conference on Image Processing*, 2020, pp. 370–374.
- [76] L. Wald, T. Ranchin, M. Mangolini, Fusion of satellite images of different spatial resolutions: Assessing the quality of resulting images, *Photogrammetric Engineering and Remote Sensing* 63 (1997) 691–699.
- [77] G. Masi, D. Cozzolino, L. Verdoliva, G. Scarpa, Pansharpening by convolutional neural networks, *Remote Sensing* 8 (2016) 594.
- [78] L.-J. Deng, G. Vivone, C. Jin, J. Chanussot, Detail injection-based deep convolutional neural networks for pansharpening, *IEEE Transactions on Geoscience and Remote Sensing* (2020).
- [79] X. Fu, W. Wang, Y. Huang, X. Ding, J. Paisley, Deep multiscale detail networks for multiband spectral image sharpening, *IEEE Transactions on Neural Networks and Learning Systems* 32 (2021) 2090–2104.
- [80] G. Scarpa, S. Vitale, D. Cozzolino, Target-adaptive cnn-based pansharpening, *IEEE Transactions on Geoscience and Remote Sensing* 56 (2018) 5443–5457.
- [81] C. Liu, Y. Zhang, S. Wang, M. Sun, Y. Ou, Y. Wan, X. Liu, Band-independent encoder–decoder network for pan-sharpening of remote sensing images, *IEEE Transactions on Geoscience and Remote Sensing* 58 (2020) 5208–5223.
- [82] J. Hu, P. Hu, X. Kang, H. Zhang, S. Fan, Pan-sharpening via multiscale dynamic convolutional neural network, *IEEE Transactions on Geoscience and Remote Sensing* 59 (2021) 2231–2244.
- [83] J. Yang, X. Fu, Y. Hu, Y. Huang, X. Ding, J. Paisley, Pannet: A deep network architecture for pan-sharpening, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 5449–5457.
- [84] H. Xu, J. Ma, Z. Shao, H. Zhang, J. Jiang, X. Guo, Sdpnet: A deep network for pan-sharpening with enhanced information representation, *IEEE Transactions on Geoscience and Remote Sensing* 59 (2021) 4120–4134.
- [85] Y. Zhang, C. Liu, M. Sun, Y. Ou, Pan-sharpening using an efficient bidirectional pyramid network, *IEEE Transactions on Geoscience and Remote Sensing* 57 (2019) 5549–5563.
- [86] H. Shen, M. Jiang, J. Li, Q. Yuan, Y. Wei, L. Zhang, Spatial–spectral fusion by combining deep learning and variational model, *IEEE Transactions on Geoscience and Remote Sensing* 57 (2019) 6169–6181.
- [87] H. Zhang, J. Ma, Gtp-pnet: A residual learning network based on gradient transformation prior for pansharpening, *ISPRS Journal of Photogrammetry and Remote Sensing* 172 (2021) 223–239.
- [88] S. Luo, S. Zhou, Y. Feng, J. Xie, Pansharpening via unsupervised convolutional neural networks, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 13 (2020) 4295–4310.
- [89] Q. Liu, H. Zhou, Q. Xu, X. Liu, Y. Wang, Psgan: A generative adversarial network for remote sensing image pan-sharpening, *IEEE Transactions on Geoscience and Remote Sensing* (2020).
- [90] Z. Shao, Z. Lu, M. Ran, L. Fang, J. Zhou, Y. Zhang, Residual encoder–decoder conditional generative adver-

- serial network for pansharpening, *IEEE Geoscience and Remote Sensing Letters* 17 (2019) 1573–1577.
- [91] J. Ma, W. Yu, C. Chen, P. Liang, X. Guo, J. Jiang, Pan-gan: An unsupervised pan-sharpening method for remote sensing image fusion, *Information Fusion* 62 (2020) 110–120.
- [92] X. Zhang, W. Huang, Q. Wang, X. Li, Ssr-net: Spatial-spectral reconstruction network for hyperspectral and multispectral image fusion, *IEEE Transactions on Geoscience and Remote Sensing* (2020).
- [93] Q. Xie, Z. Minghao, Q. Zhao, Z. Xu, D. Meng, Mhf-net: An interpretable deep network for multispectral and hyperspectral image fusion, *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2020).
- [94] S. Xu, O. Amira, J. Liu, C.-X. Zhang, J. Zhang, G. Li, Ham-mfn: Hyperspectral and multispectral image multiscale fusion network with rap loss, *IEEE Transactions on Geoscience and Remote Sensing* 58 (2020) 4618–4628.
- [95] W. Wei, J. Nie, Y. Li, L. Zhang, Y. Zhang, Deep recursive network for hyperspectral image super-resolution, *IEEE Transactions on Computational Imaging* 6 (2020) 1233–1244.
- [96] T. Uezato, D. Hong, N. Yokoya, W. He, Guided deep decoder: Unsupervised image pair fusion, in: *Proceedings of the European Conference on Computer Vision*, 2020, pp. 87–102.
- [97] Y. Qu, H. Qi, C. Kwan, Unsupervised sparse dirichlet-net for hyperspectral image super-resolution, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 2511–2520.
- [98] R. Dian, S. Li, X. Kang, Regularizing hyperspectral and multispectral image fusion by cnn denoiser, *IEEE Transactions on Neural Networks and Learning Systems* 32 (2021) 1124–1135.
- [99] J. Yao, D. Hong, J. Chanussot, D. Meng, X. Zhu, Z. Xu, Cross-attention in coupled unmixing nets for unsupervised hyperspectral super-resolution, in: *Proceedings of the European Conference on Computer Vision*, 2020, pp. 208–224.
- [100] K. Zheng, L. Gao, W. Liao, D. Hong, B. Zhang, X. Cui, J. Chanussot, Coupled convolutional neural network with adaptive response function learning for unsupervised hyperspectral super resolution, *IEEE Transactions on Geoscience and Remote Sensing* 59 (2021) 2487–2502.
- [101] W. Wang, W. Zeng, Y. Huang, X. Ding, J. Paisley, Deep blind hyperspectral image fusion, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2019, pp. 4150–4159.
- [102] W. Xie, Y. Cui, Y. Li, J. Lei, Q. Du, J. Li, Hpgan: Hyperspectral pansharpening using 3-d generative adversarial networks, *IEEE Transactions on Geoscience and Remote Sensing* 59 (2021) 463–477.
- [103] F. Zhao, et al., Learning specific and general realm feature representations for image fusion, *IEEE Transactions on Multimedia* (2020).
- [104] X. Deng, P. L. Dragotti, Deep convolutional neural network for multi-modal image restoration and fusion, *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2020).
- [105] H. Jung, Y. Kim, H. Jang, N. Ha, K. Sohn, Unsupervised deep image fusion with structure tensor representations, *IEEE Transactions on Image Processing* 29 (2020) 3845–3858.

- [106] H. Xu, J. Ma, Emfusion: An unsupervised enhanced medical image fusion network, *Information Fusion* 76 (2021) 177–186.
- [107] J. Cai, S. Gu, L. Zhang, Learning a deep single image contrast enhancer from multi-exposure images, *IEEE Transactions on Image Processing* 27 (2018) 2049–2062.
- [108] M. Nejati, S. Samavi, S. Shirani, Multi-focus image fusion using dictionary-based sparse representation, *Information Fusion* 25 (2015) 72–84.
- [109] F. Yasuma, T. Mitsunaga, D. Iso, S. K. Nayar, Generalized assorted pixel camera: postcapture control of resolution, dynamic range, and spectrum, *IEEE Transactions on Image Processing* 19 (2010) 2241–2253.
- [110] C.-H. Lee, L.-H. Chen, W.-K. Wang, Image contrast enhancement using classified virtual exposure image fusion, *IEEE Transactions on Consumer Electronics* 58 (2012) 1253–1261.
- [111] Z. Wang, Y. Ma, J. Gu, Multi-focus image fusion using pcnn, *Pattern Recognition* 43 (2010) 2003–2016.
- [112] O. Bimber, D. Iwai, Superimposing dynamic range, *ACM Transactions on Graphics* 27 (2008) 1–8.
- [113] S. Kuthirummal, H. Nagahara, C. Zhou, S. K. Nayar, Flexible depth of field photography, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 33 (2010) 58–71.
- [114] S. Liu, Y. Zhang, Detail-preserving underexposed image enhancement via optimal weighted multi-exposure fusion, *IEEE Transactions on Consumer Electronics* 65 (2019) 303–311.
- [115] Y. Zhang, X. Bai, T. Wang, Boundary finding based multi-focus image fusion through multi-scale morphological focus-measure, *Information Fusion* 35 (2017) 81–101.
- [116] K. Jędrasiak, A. Nawrat, K. Daniec, R. Koterias, M. Mikulski, T. Grzejszczak, A prototype device for concealed weapon detection using ir and cmos cameras fast image fusion, in: *Proceedings of the International Conference on Computer Vision and Graphics*, 2012, pp. 423–432.
- [117] Y. Kinoshita, S. Shiota, H. Kiya, Automatic exposure compensation for multi-exposure image fusion, in: *Proceedings of the IEEE International Conference on Image Processing*, 2018, pp. 883–887.
- [118] X. Zhang, P. Ye, H. Leung, K. Gong, G. Xiao, Object fusion tracking based on visible and infrared images: A comprehensive review, *Information Fusion* 63 (2020) 166–187.
- [119] H. Liu, F. Sun, Fusion tracking in color and infrared images using sequential belief propagation, in: *Proceedings of the IEEE International Conference on Robotics and Automation*, 2008, pp. 2259–2264.
- [120] S. R. Schnelle, A. L. Chan, Enhanced target tracking through infrared-visible image fusion, in: *Proceedings of the International Conference on Information Fusion*, 2011, pp. 1–8.
- [121] X. Mei, H. Ling, Robust visual tracking using l1 minimization, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2009, pp. 1436–1443.
- [122] Y. Wu, E. Blasch, G. Chen, L. Bai, H. Ling, Multiple source data fusion via sparse representation for robust visual tracking, in: *Proceedings of the IEEE International Conference on Information Fusion*, 2011, pp. 1–8.
- [123] C. Li, C. Zhu, Y. Huang, J. Tang, L. Wang, Cross-modal ranking with soft consistency and noisy labels for

- robust rgb-t tracking, in: Proceedings of the European Conference on Computer Vision, 2018, pp. 808–823.
- [124] C. Li, C. Zhu, J. Zhang, B. Luo, X. Wu, J. Tang, Learning local-global multi-graph descriptors for rgb-t object tracking, *IEEE Transactions on Circuits and Systems for Video Technology* 29 (2018) 2913–2926.
- [125] M. Wang, Y. Liu, Z. Huang, Large margin object tracking with circulant feature maps, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 4021–4029.
- [126] S. Zhai, P. Shao, X. Liang, X. Wang, Fast rgb-t tracking via cross-modal correlation filters, *Neurocomputing* 334 (2019) 172–181.
- [127] K. Zhang, Q. Liu, Y. Wu, M.-H. Yang, Robust visual tracking via convolutional networks without training, *IEEE Transactions on Image Processing* 25 (2016) 1779–1792.
- [128] X. Lan, M. Ye, R. Shao, B. Zhong, P. C. Yuen, H. Zhou, Learning modality-consistency feature templates: A robust rgb-infrared tracking system, *IEEE Transactions on Industrial Electronics* 66 (2019) 9887–9897.
- [129] X. Lan, M. Ye, S. Zhang, P. Yuen, Robust collaborative discriminative learning for rgb-infrared tracking, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2018, pp. 7008–7015.
- [130] C. Li, H. Cheng, S. Hu, X. Liu, J. Tang, L. Lin, Learning collaborative sparse representation for grayscale-thermal tracking, *IEEE Transactions on Image Processing* 25 (2016) 5743–5756.
- [131] C. Li, N. Zhao, Y. Lu, C. Zhu, J. Tang, Weighted sparse representation regularized graph learning for rgb-t object tracking, in: Proceedings of the ACM International Conference on Multimedia, 2017, pp. 1856–1864.
- [132] S. L. Wellington, H. J. Vinegar, et al., X-ray computerized tomography, *Journal of Petroleum Technology* 39 (1987) 885–898.
- [133] C. Degen, M. Poggio, H. Mamin, C. Rettner, D. Rugar, Nanoscale magnetic resonance imaging, *Proceedings of the National Academy of Sciences* 106 (2009) 1313–1317.
- [134] S. S. Gambhir, Molecular imaging of cancer with positron emission tomography, *Nature Reviews Cancer* 2 (2002) 683–693.
- [135] M. Horger, R. Bares, The role of single-photon emission computed tomography/computed tomography in benign and malignant bone disease, *Seminars in Nuclear Medicine* 36 (2006) 286–294.
- [136] A. Achim, A. Bezerianos, P. Tsakalides, Novel bayesian multiscale method for speckle removal in medical ultrasound images, *IEEE Transactions on Medical Imaging* 20 (2001) 772–783.
- [137] B. Rajalingam, R. Priya, Hybrid multimodality medical image fusion technique for feature enhancement in medical diagnosis, *International Journal of Engineering Science Invention* 2 (2018) 52–60.
- [138] Y. Li, J. Zhao, Z. Lv, J. Li, Medical image fusion method by deep learning, *International Journal of Cognitive Computing in Engineering* 2 (2021) 21–29.
- [139] C. Li, A. Zhu, Application of image fusion in diagnosis and treatment of liver cancer, *Applied Sciences* 10 (2020) 1171.
- [140] E. Xu, Y. Long, K. Li, Q. Zeng, L. Tan, L. Luo, Q. Huang, R. Zheng, Comparison of ct/mri-ceus and us-ceus

fusion imaging techniques in the assessment of the thermal ablation of liver tumors, *International Journal of Hyperthermia* 35 (2018) 159–167.

[141] M. W. Lee, Fusion imaging of real-time ultrasonography with ct or mri for hepatic intervention, *Ultrasonography* 33 (2014) 227.

[142] A. A. Pure, N. Gupta, M. Shrivastava, An overview of different image fusion methods for medical applications, *International Journal of Scientific and Engineering Research* 4 (2013) 129–133.

[143] Z. Shao, H. Fu, D. Li, O. Altan, T. Cheng, Remote sensing monitoring of multi-scale watersheds impermeability for urban hydrological evaluation, *Remote Sensing of Environment* 232 (2019) 111338.

[144] N. Rudorff, C. M. Rudorff, M. Kampel, G. Ortiz, Remote sensing monitoring of the impact of a major mining wastewater disaster on the turbidity of the doce river plume off the eastern brazilian coast, *ISPRS Journal of Photogrammetry and Remote Sensing* 145 (2018) 349–361.

[145] R. R. Colditz, T. Wehrmann, M. Bachmann, K. Steinnocher, M. Schmidt, G. Strunz, S. Dech, Influence of image fusion approaches on classification accuracy: a case study, *International Journal of Remote Sensing* 27 (2006) 3311–3335.

[146] H. Zhang, J. Ma, C. Chen, X. Tian, Ndv-net: A fusion network for generating high-resolution normalized difference vegetation index in remote sensing, *ISPRS Journal of Photogrammetry and Remote Sensing* 168 (2020) 182–196.

[147] S. Rajalakshmi, V. V. Chamundeeswari, Mapping of mineral deposits using image fusion by pca approach, in: *Proceedings of IEEE International Conference on Computer Communication and Systems*, 2014, pp. 024–029.

[148] X. Fu, Z. Lin, Y. Huang, X. Ding, A variational pan-sharpening with local gradient constraints, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2019, pp. 10265–10274.

[149] C. Pohl, J. van Genderen, Remote sensing image fusion: an update in the context of digital earth, *International Journal of Digital Earth* 7 (2014) 158–172.

[150] J. Ma, X. Jiang, A. Fan, J. Jiang, J. Yan, Image matching from handcrafted to deep features: A survey, *International Journal of Computer Vision* 129 (2021) 23–79.

[151] X. Jiang, J. Ma, G. Xiao, Z. Shao, X. Guo, A review of multimodal image matching: Methods and applications, *Information Fusion* 73 (2021) 22–71.

[152] H. Li, Y. Cen, Y. Liu, X. Chen, Z. Yu, Different input resolutions and arbitrary output resolution: A meta learning-based deep framework for infrared and visible image fusion, *IEEE Transactions on Image Processing* 30 (2021) 4070–4083.